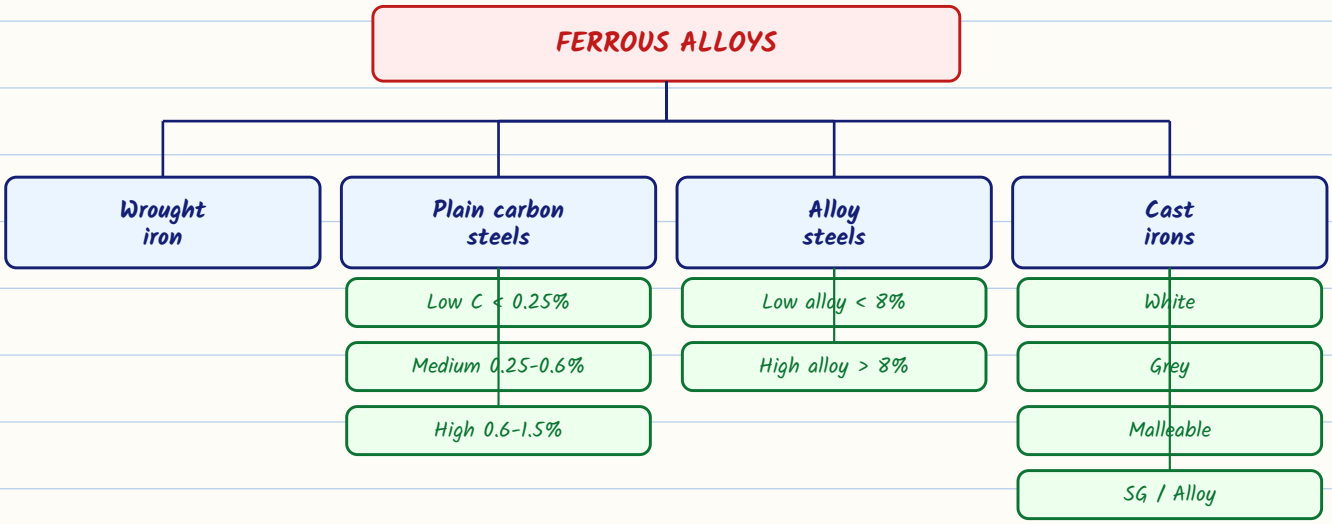
UNIT - II : FERROUS METALS AND ALLOYS

1. Introduction and classification

* Ferrous metals have IRON as the base: steels and cast irons. Iron is the cheapest and most used engineering metal - about 90% of all metal tonnage.
* Carbon content decides the family: Wrought iron < 0.08% C, Steels 0.08 - 2.14% C, Cast irons 2.14 - 6.67% C (usually 2.5 - 4% C with 1 - 3% Si).



2. Cast Irons

* Cast iron = Fe - C - Si alloy with C above 2.14% (in practice 2.5 - 4.0%), Si 1 - 3%, Mn, S, P in small amounts. Good fluidity, low melting point (1150 - 1300 C), easy casting, excellent damping, wear resistance and machinability; but low ductility.
* Carbon appears either as CEMENTITE (Fe3C - white iron) or as free

GRAPHITE (grey, malleable, SG). Factors: cooling rate, composition (Si, a graphitiser), section thickness.

Si promotes graphite; S, Mn, Cr and fast cooling promote cementite (white iron). 'Greener' graphite -> softer, more machinable iron.

(a) White cast iron

* All carbon is combined as Fe3C. Produced by rapid cooling (chilling) of low Si (0.5 - 1.3%) melt; fracture is bright white and crystalline.
* Structure: pearlite + cementite + (transformed) ledeburite. Very hard (350-500 BHN), very brittle, highly wear and abrasion resistant, not machinable (only grinding), tensile strength about 200-400 MPa, compression strong.
* Uses: crushing and grinding balls, rolls, liners, ploughshares, rail brake shoes, extrusion nozzles, and as the parent material for malleable iron. Chilled iron castings have white iron surface over grey core (car wheels, rolls, cams).

(b) Grey cast iron

* Composition: 2.5-4% C, 1-3% Si, 0.4-1% Mn, S < 0.15%, P < 0.3%. Carbon is present as FLAKES of graphite in a matrix of ferrite, pearlite or both. Fracture appears grey.
* Slow cooling and high Si produce graphite flakes which act as internal notches -> low tensile strength (150 - 400 MPa, grades FG 150 to FG 400), almost no ductility (< 1%), but compressive strength 3 - 4 times the tensile.
* Properties: excellent machinability, castability, wear resistance (graphite is a solid lubricant), very good vibration DAMPING and resistance to compressive loads; low cost; poor impact strength.

* Uses: machine tool beds, engine blocks, cylinder heads and liners, gear housings, pipes, pulleys, brake drums, manhole covers, flywheels.

(c) Malleable cast iron

* White iron (C 2.2-2.9%, Si 0.9-1.9%) given long MALLEABILISING heat treatment (annealing) which breaks down cementite: Fe3C -> 3Fe + C (temper carbon, compact rosettes).

Type	Process	Structure	Property
Whiteheart	Heat 900-1050 C in oxidising medium (haematite ore), 60-100 h	Ferrite + temper carbon, decarburised skin	Less ductile, weldable
Blackheart	Stage I 900-950 C (48 h) + slow cool to 700 C, neutral atm.	Ferrite matrix + nodules of temper carbon	Ductile, good machinability
Pearlitic	Cool faster, then 2nd stage short	Pearlite + temper carbon	Stronger, wear resistant

* Tensile strength 300-500 MPa, elongation 6 - 14%, good shock resistance and machinability. Uses: automobile parts (crankshafts, connecting rods, differential housings), pipe fittings, hand tools, agricultural implements, railroad castings. Limited to thin sections (< 40 mm).

(d) Spheroidal graphite (SG / ductile / nodular) cast iron

* Graphite appears as SPHEROIDS (nodules). Obtained by adding 0.03 - 0.06% MAGNESIUM (or Ce) to the melt (after desulphurising) followed by inoculation with ferro-silicon (0.5-1%).
* Sphere shape removes stress concentration -> high strength (400 - 800 MPa) AND ductility (2 - 18%). Matrix may be ferritic (ductile),

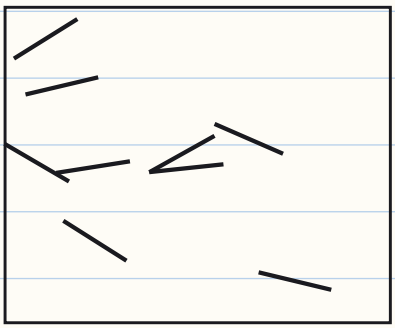
pearlitic (strong) or bainitic / martensitic (ADI: austempered ductile iron).

* Uses: crankshafts, pistons, gears, pipes (water, sewage), dies, valves, hydraulic parts, wind turbine hubs, automobile suspension parts, steel mill rolls.

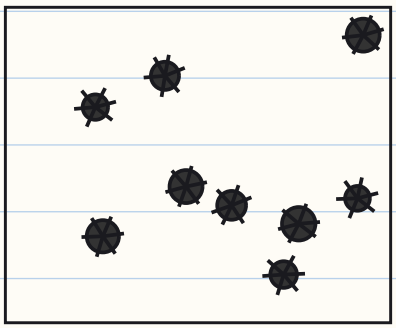
(e) Alloy cast irons

* Contain special alloying elements to give high strength, wear, heat and corrosion resistance.
* Ni-hard (4% Ni, 2% Cr): martensitic white iron for abrasion. Ni-resist (14-30% Ni): austenitic, corrosion and heat resistant. Silal / high-silicon (14-17% Si): acid resistant (Duriron). High Cr iron (12-35% Cr): heat and abrasion resistant (pump casings, furnace parts). Mo and Cu increase strength of grey iron.

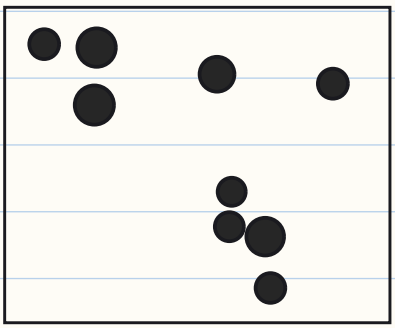
Property	White	Grey	Malleable	SG iron
Graphite	None	Flakes	Rosettes	Spheres
Tensile MPa	200-400	150-400	300-500	400-800
Ductility %	~0	< 1	6-14	2-18
Machinability	Poor	Excellent	Good	Good



Grey iron: FLAKES



Malleable: ROSETTES



SG iron: SPHEROIDS

Effect of elements on cast iron

* CARBON: increases graphite, lowers melting point and strength.

SILICON: strong graphitiser (2-3%), improves fluidity and acid resistance. MANGANESE: opposes graphitisation (carbide former), neutralises S, adds strength (0.5-1%).

* SULPHUR: carbide stabiliser, causes hot shortness, reduces fluidity (keep < 0.1%). PHOSPHORUS: forms hard, brittle steadite (Fe-Fe3C-Fe3P eutectic, m.p. 950 C), increases fluidity (thin castings, ornamental).
* Carbon equivalent CE = %C + (%Si + %P)/3 ; CE = 4.3 is eutectic composition. Higher CE gives lower strength grey iron.

Cast iron vs steel

Cast iron	Steel
2.14 - 6.67% C, Si 1-3%	0.05 - 2.14% C, Si < 0.6%
Cast only (brittle), hardly forgeable	Can be rolled, forged, welded, heat treated
Low cost, damping, wear resistance	Higher tensile strength and toughness
Melting point 1150-1300 C	Melting point 1370-1530 C

3. Steels

Classification of steels

* By CARBON: Low carbon / mild steel (0.05 - 0.25% C), medium carbon (0.25 - 0.6%), high carbon (0.6 - 1.5%).
* By ALLOYING: plain carbon steels (only C, Mn <= 1.65%, Si <= 0.6%), low alloy (total alloy < 8%), high alloy (> 8%: stainless, tool steels).
* By DEOXIDATION: killed, semi-killed, rimmed steel. By application: structural, tool, spring, stainless, high speed.

* By structure: ferritic, pearlitic, austenitic, martensitic, bainitic. IS designation: 40C8 (0.40% C, 0.8% Mn), 20CrlMo28.

Plain carbon steels – structure and properties

* As carbon rises: hardness, tensile strength, wear resistance increase; ductility, toughness, weldability and machinability decrease. Structure follows Fe-Fe3C diagram: ferrite + pearlite (hypo), pearlite (0.77%), pearlite + Fe3C network (hyper).

Type	%C	Structure	Properties	Uses
Dead mild	0.05-0.15	Mostly ferrite	Very ductile, weldable, UTS 300-400	Wires, sheets, rivets, pipes, car bodies
Mild	0.15-0.3	Ferrite + little pearlite	UTS 400-500, tough	Shafts, bolts, structural, case hardening
Medium	0.3-0.6	Ferrite + pearlite	UTS 550-800, heat treatable	Axles, gears, crankshafts, rails
High	0.6-1.0	Mostly pearlite	Hard, strong, less ductile	Springs, hammers, dies, wire ropes
Very high	1.0-1.5	Pearlite + Fe3C	Very hard, wear resist., brittle	Files, razors, drills, cutters

* Effect of impurities: Mn (deoxidiser, forms MnS, strength); Si (deoxidiser, strength); S (hot shortness - forms FeS, so Mn added); P (cold shortness - brittleness, but improves machinability); O, N, H harmful.

Low alloy steels

* Alloy additions < 8% (Ni, Cr, Mo, V, Mn, Si, W) to increase hardenability, strength, toughness, and low temperature / corrosion

properties without losing weldability.

* HSLA (high strength low alloy) steels: < 0.2% C with Cu, V, Nb, Ti (micro-alloys); yield 300-550 MPa; used in bridges, pipelines, car frames, pressure vessels.
* Examples: 1-3% Ni steel (toughness; gears), Ni-Cr steel (shafts, gears), Cr-Mo steels (pressure vessels, high temp.), Cr-V steel (springs, tools), Si-Mn steel (springs - 60Si2Mn).

Hadfield manganese steel

* Composition: 1.0 - 1.4% C and 10 - 14% Mn (about 12% Mn) - AUSTENITIC steel. After water quenching from 1000 - 1100 C (toughening treatment) the structure is fully tough austenite.
* Special property: WORK HARDENING - the surface becomes extremely hard (500+ BHN) under impact or abrasion while the core remains tough and ductile. Non-magnetic, very poor machinability (use grinding), difficult to weld.
* Uses: rock crusher jaws, rail-road points and crossings, excavator bucket teeth, dredger parts, grinding mill liners, bullet-proof plates, safes.

Effect of alloying elements in steels

Element	Effect on steel
Nickel (Ni)	Stabilises austenite, increases toughness, strength and low-temperature impact; corrosion resistance
Chromium (Cr)	Carbide former, hardenability, wear, corrosion and oxidation resistance; stabilises ferrite
Molybdenum (Mo)	Hardenability, high temperature strength, creep resistance; reduces temper brittleness
Vanadium (V)	Strong carbide former, fine grain, secondary hardening, toughness
Tungsten (W)	Hot hardness and red hardness, wear resistance (HSS)
Manganese (Mn)	Hardenability, strength; at 12% gives work hardening austenite
Silicon (Si)	Deoxidiser, strengthens ferrite, magnetic and spring properties
Cobalt (Co)	Increases red hardness; permanent magnets
Titanium / Niobium	Strong carbide formers; stabilise stainless steel against intergranular corrosion

Stainless steels

* Minimum 10.5 - 12% Cr forms thin passive Cr2O3 film. Ferritic (12-27% Cr, magnetic), Martensitic (12-18% Cr, 0.1-1% C, hardenable: cutlery, turbine blades), Austenitic (18% Cr - 8% Ni, '18/8', non-magnetic, excellent corrosion resistance: kitchenware, chemical plant).

Type	Typical grade	Structure	Remarks
Ferritic	430: 17Cr, 0.1C	BCC ferrite, magnetic	Not hardenable; decorative, automobile trim
Martensitic	410: 12Cr, 0.15C; 420: 13Cr, 0.35C	Martensite after quench	Hardenable; cutlery, turbine blades, surgical tools
Austenitic	304: 18Cr-8Ni; 316: + 2-3% Mo	FCC, non magnetic	Best corrosion, weldable; not hardenable by heat treatment
Precipitation hardening	17-4 PH, 17-7 PH	Martensite + Cu or Al ppts	High strength aerospace
Duplex	2205: 22Cr-5Ni	Ferrite + austenite	Stress corrosion resistant

* Sensitisation / weld decay: in 18/8 steel heated at 500-800 C chromium carbide ($Cr_{23}C_6$) forms at grain boundaries, depleting Cr -> intergranular corrosion. Prevented by low C (304L), stabilising with Ti / Nb (321, 347) or solution annealing at 1050 C.

Other special steels

* FREE CUTTING steels: added S (0.1-0.3%), Pb or P -> easy chip breaking, used for screws, bolts made on automatic machines (e.g. 11SMn30).
* SPRING steels: 0.5-1.0% C with Si-Mn or Cr-V; high elastic limit and fatigue strength (leaf and coil springs; 55Si2Mn90, 50CrIV23).
* MARAGING steels: 18% Ni, Co, Mo, Ti with very low C; martensite + age hardening at 480 C gives UTS up to 2400 MPa and good toughness (rocket motor cases, dies).
* HEAT RESISTING steels: high Cr, Ni, Si (e.g. 25Cr-20Ni) for furnace parts; BEARING steel: 1% C - 1.5% Cr (EN31) for ball and roller

bearings; electrical (Si) steel for transformers.

Tool and die steels

* Used for making cutting tools, dies and punches; need high hardness (60-67 HRC), wear resistance, red hardness (hot hardness), toughness and dimensional stability. Always heat treated (harden + temper).

Class	Typical composition	Properties and uses
Plain carbon tool steel	0.6-1.5% C	Water hardening, loses hardness above 250 C; files, chisels, punches
High speed steel (HSS) 18-4-1	0.7C - 18W - 4Cr - 1V	Red hardness to 600 C; drills, milling cutters, lathe tools
Mo-HSS 6-6-4-2	6W-5Mo-4Cr-2V	Cheaper, tougher HSS; taps, reamers
Hot work die steel (H-series)	5% Cr, Mo, V, W	Resist thermal fatigue; forging / extrusion / die casting dies
Cold work (D-series)	1.5C - 12Cr - Mo	High wear, minimum distortion; blanking dies, gauges
Shock resisting (S)	0.5C - Si - Cr - W	Toughness; chisels, rivet sets, punches

Rule: Cutting tool materials in order of hot hardness: carbon steel < HSS < cemented carbides < ceramics < diamond.

NON-FERROUS METALS AND ALLOYS

* All metals other than iron-base: lighter, better corrosion resistance, high electrical / thermal conductivity, easy fabrication, special

properties; costlier than steel.

4. Copper and its alloys

* Pure Cu: FCC, density 8.96 g/cc, m.p. 1085 C, very high electrical (second to Ag) and thermal conductivity, ductile, corrosion resistant, non-magnetic. Uses: wires, cables, busbars, heat exchangers, roofing. Types: ETP copper (99.9%, 0.04% O), OFHC (oxygen free high conductivity) and deoxidised copper.

Alloy	Composition	Properties	Uses
Cartridge brass	70 Cu - 30 Zn	Alpha brass; best ductility, deep drawing	Cartridge cases, radiators
Muntz metal	60 Cu - 40 Zn	Alpha + beta; hot workable, strong	Condenser tubes, bolts, ship sheathing
Gilding metal	95 Cu - 5 Zn	Colour of gold	Jewellery, coins, medals
Admiralty brass	70 Cu-29 Zn-1 Sn	Sea water resistant	Condenser tubes
Tin bronze (gun metal)	88 Cu-10 Sn-2 Zn	Strong, wear resistant	Bearings, gears, valves, guns
Phosphor bronze	Cu-4-10 Sn-0.3 P	Fatigue resistant, spring	Springs, bearings, gears
Aluminium bronze	Cu-5-11 Al	High strength, corrosion	Propellers, pump parts
Beryllium bronze	Cu-2 Be	Age hardenable, 1400 MPa	Springs, non-sparking tools
Cupro-nickel / Monel	Cu-30 Ni	Corrosion resistance	Coins, condenser tubes

* BRASS = Cu + Zn (up to 40-45%), BRONZE = Cu + Sn (or Al, Si, Be). Alpha brass (< 36% Zn): single phase FCC, cold workable; alpha + beta brass (36 - 45% Zn): stronger but harder to cold work. Season cracking and dezincification are typical defects of brass.

5. Aluminium and its alloys

* Pure Al: FCC, density 2.7 g/cc (1/3 of steel), m.p. 660 C, excellent corrosion resistance (Al2O3 film), electrical conductivity 62% of Cu, very ductile, non-magnetic, non-toxic, easily recyclable. Low strength (UTS ~ 90 MPa) - improved by alloying, cold work and age hardening.
* Wrought alloys (designation 1xxx to 8xxx): 1xxx pure Al, 2xxx Cu (duralumin), 3xxx Mn, 4xxx Si, 5xxx Mg, 6xxx Mg + Si, 7xxx Zn (highest strength), 8xxx others.

Alloy	Composition	Notes / Uses
Duralumin (2017)	4 Cu - 0.5 Mg - 0.7 Mn - Al	Age hardenable; aircraft structure, rivets, automobile
Y-alloy	4 Cu - 2 Ni - 1.5 Mg	Retains strength at high T; pistons, cylinder heads
Hindalium	Al-Mg-Mn-Cr	Anodised; utensils
Magnalium	2-10 Mg - Al	Light weight; aircraft parts, balances
Al-Si alloy (Silumin)	10-13 Si	Eutectic; excellent castability: pistons, engine blocks
7075	5.6 Zn-2.5 Mg-1.6 Cu	UTS up to 570 MPa; aircraft wings
6061	Mg + Si	Weldable, extruded sections, frames

* Age hardening of duralumin: solution treat at 500 C, quench, age at room temp. (natural) or 150 C (artificial) -> GP zones / theta'

(CuAl2) precipitates.

6. Titanium and its alloys

* Pure Ti: HCP alpha below 882 C, BCC beta above; density 4.5 g/cc, m.p. 1668 C, high strength-to-weight ratio, outstanding corrosion resistance (TiO2 film), low thermal expansion; costly (Kroll process); poor machinability; reacts with O, N, H at high T so melted in vacuum / argon.
* ALPHA alloys (Ti-5Al-2.5Sn): weldable, creep resistant, not heat treatable. BETA alloys (Ti-13V-11Cr-3Al): hardenable, formable. ALPHA-BETA alloys (Ti-6Al-4V, 'workhorse', 50% of all Ti use): heat treatable, UTS ~ 1000 MPa.
* Uses: aircraft engine compressor blades, airframes, missile parts, chemical plant, surgical implants (bio-compatible), hip joints, marine parts, sports goods.

7. Magnesium and its alloys

* Mg: HCP, density 1.74 g/cc (lightest structural metal), m.p. 650 C, high damping, low modulus, poor corrosion and fatigue resistance, burns in air if finely divided, poor formability at room temperature (HCP: only 3 slip systems) - worked hot.
* Alloys: Mg-Al-Zn (AZ31 wrought, AZ91 die casting), Mg-Zn-Zr (ZK60), Mg-Th (high temperature), Elektron (Mg-Al-Zn), Mg-Li (ultra light). Uses: aircraft and automobile parts (wheels, gear box housings), laptops and phone bodies, portable tools, reflectors, sacrificial anodes.

8. Super alloys

* Alloys (Ni, Co or Fe base) with excellent strength, creep, fatigue, oxidation and corrosion resistance at VERY HIGH temperatures (up to

0.8-0.9 T_m, > 1000 C). They are for gas turbine and jet engine parts.

* Nickel base (Inconel, Nimonic, Waspaloy, Udimet, Hastelloy): matrix gamma (FCC Ni solid solution) strengthened by gamma-prime precipitates Ni3(Al,Ti) (coherent) + carbides (M23C6, MC) at grain boundaries.
* Cobalt base (Stellite, Haynes 25, Vitallium): solid solution + carbide strengthening; good hot corrosion and wear resistance; used in vanes, bio-implants.
* Iron-nickel base (A-286, Incoloy 800): cheaper, austenitic stainless type + gamma-prime. Strengthening mechanisms: solid solution, precipitation, dispersion (ODS - oxide dispersion strengthened), directional solidification and SINGLE CRYSTAL blades (removing grain boundaries stops creep).
* Uses: turbine blades and discs, combustion chambers, afterburners, nuclear reactor parts, heat exchangers, rocket engine nozzles, chemical reactors.

Comparison of light metals

Metal	Density g/cc	M.P. C	Structure	UTS (alloyed) MPa	Key use
Aluminium	2.7	660	FCC	200-570	Aircraft, packaging, cables
Magnesium	1.74	650	HCP	200-350	Lightweight casings
Titanium	4.5	1668	HCP/BCC	900-1200	Jet engines, implants
Copper	8.96	1085	FCC	250-1400 (Be-Cu)	Electrical, bearings
Nickel SA	8.2-8.9	1350	FCC	1000-1400	Turbine blades

Important points for exams

* 1. Distinguish white, grey, malleable and SG cast irons (graphite form, process, properties, uses).
* 2. Classify steels; explain properties of plain carbon steel in relation to % C.
* 3. Hadfield manganese steel: composition, toughening heat treatment, work hardening, uses.
* 4. Compositions and applications of brass, bronze, duralumin, Ti-6Al-4V, Mg alloys and super alloys.
* 5. Why are super alloys used in gas turbines? (gamma-prime strengthening, creep resistance, oxidation resistance, single crystal blades).

Revision: Hadfield steel = 12% Mn work hardening; Nimonic = Ni-Cr-Ti-Al super alloy; Duralumin = Al-Cu age hardening; Ti-6Al-4V = aerospace; SG iron = Mg added; Malleable iron = annealed white iron.